

Konstantinos Kogkalidis
CURRICULUM VITAE

| Structure-Aware Machine Learning | (AI for) Type & Proof Theory |
| Neurosymbolic Integration | Computational Linguistics |

Updated August 19, 2026.

Personalia

Date of Birth 19 July 1991
Nationality Greek
Current Residence Thessaloniki, Greece
Pronouns he/him
Email [konstantinos at riseup dot net](mailto:konstantinos@riseup.net)
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Positions

09/2026 - today **Scientific Consultant**
Affiliation [Kodamai](#)

09/2025 - 08/2026 **Senior AI Specialist**
Affiliation [Input Output \(IOG\)](#)

07/2023 - 07/2025 **Postdoctoral Researcher**
Affiliation Department of Computer Science, Aalto University

09/2024 - 02/2025 **Visiting Researcher**
Affiliation Department of Linguistics, University of Bologna

02/2024 - 12/2024 **ML Research Scientist**
Affiliation [Skyfora](#)

09/2018 - 12/2022 **Doctoral Researcher**
Affiliation Utrecht Institute of Linguistics, Utrecht University

07/2017 - 09/2018 **ML Engineer**
Affiliation [Geekbot](#)

Education

09/2018 - 12/2022 **PhD Candidate** in *Computational Linguistics*
Institution Utrecht Institute of Linguistics, Utrecht University
Thesis [Dependency as Modality, Parsing as Permutation](#)
Promoters Michael Moortgat & Richard Moot
Committee A. v. d. Bosch, M. Sadrzadeh, J. Bos, S. Clark, S. Chatzikyriakidis
Award [E. W. Beth Dissertation Prize, FoLLI \(2024\)](#)

09/2017 - 07/2019 **Research MSc** in *Artificial Intelligence* (120 EC, cum laude)
Track Logic & Reasoning
Institution Graduate School of Natural Sciences, Utrecht University

11/2010 - 05/2017 **Engineering Diploma** in *Electrical & Computer Engineering* (300 EC, MSc equivalent)
Track Electronics & Computers
Institution Department of Engineering, Aristotle University of Thessaloniki

Teaching

- 2022 - 2023** Compositional Models of Vector-Based Semantics (ESSLLI 2022, guest)
2021 - 2022 Logic & Language (UU AI MSc, guest, TA)
2020 - 2021 Logic & Language (UU AI MSc, main instructor)
2019 - 2020 Machine Learning for Human Vision & Language (UU AI MSc, TA)
2018 - 2019 Logic & Language (UU AI MSc, tutorials, TA)
Advanced Topics in Cognitive Modeling (UU AI MSc, TA)

Scholarships, Grants & Awards

- 12/2024 Concurrent Programming and Lambek Calculi** (Working Visit)
Awarded Amount 1,500 €
Awarding Body Royal Netherlands Academy of Arts and Sciences ([KNAW](#))
Host Institution University of Groningen
Host Jorge Pérez
- 09/2024 Neural Premise Selection for Agda** (Short-Term Scientific Mission)
Awarded Amount 1,330 €
Awarding Body European Research Network on Formal Proofs ([EuroProofNet](#))
Host Institution Chalmers
Host Jean-Philippe Bernardy
- 07/2024 E. W. Beth Dissertation Prize** (Best Dissertation)
Awarded Amount 1,500 €
Awarding Body The Association for Logic, Language and Information ([FoLLI](#))

All my academic contributions below (papers, code, slides, etc.) are open-source and available through my website.

Publications & Preprints

PhD Thesis

[Dependency as Modality, Parsing as Permutation](#)

In *LOT Dissertation Series*. 2023.

Papers

1. [Backward through Time, Algebraically](#)
[Konstantinos Kogkalidis](#)
In *Preprint*. (2026).
2. [Nominal Class Assignment in Swahili: A Computational Account](#)
Giada Palmieri and [Konstantinos Kogkalidis](#)
In *Proceedings of the 10th Italian Conference on Computational Linguistics (CLiC-it 2024)*.
3. [On Tables with Numbers, with Numbers](#)
[Konstantinos Kogkalidis](#) and Stergios Chatzikyriakidis
In *Proceedings of the 1st Workshop on Language Models for Underserved Communities (LM4UC 2025)*.
4. [Learning Structure-Aware Representations of Dependent Types](#)
[Konstantinos Kogkalidis](#), Orestis Melkonian and Jean-Philippe Bernardy
In *Proceedings of the 38th Conference on Neural Information Processing Systems (NeurIPS 2024)*.
5. [Algebraic Positional Encodings](#)
[Konstantinos Kogkalidis](#), Jean-Philippe Bernardy and Vikas Garg
In *Proceedings of the 38th Conference on Neural Information Processing Systems (NeurIPS 2024)*.

6. **OYXOY: A Modern NLP Test Suite for Modern Greek**
Konstantinos Kogkalidis, Stergios Chatzikyriakidis, Eirini Chrysovalantou Giannikouri, Vassiliki Katsouli, Christina Klironomou, Christina Koula, Dimitris Papadakis, Thelka Pasparaki, Erofilis Psaltaki, Efthymia Sakellariou and Hara Soupiona
 In *Findings of the Association for Computational Linguistics (EACL 2024)*.
7. **SPINDLE: Spinning Raw Text into Lambda Terms with Graph Attention**
Konstantinos Kogkalidis, Michael Moortgat and Richard Moot
 In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations (EACL 2023 Demos)*.
8. **Geometry-Aware Supertagging with Heterogeneous Dynamic Convolutions**
Konstantinos Kogkalidis and Michael Moortgat
 In *Proceedings of the 2023 CLASP Conference on Learning with Small Data (LSD 2023)*.
9. **Improving BERT Pretraining with Syntactic Supervision**
 Georgios Tziafas, Konstantinos Kogkalidis, Gijs Wijnholds and Michael Moortgat
 In *Proceedings of the 2023 CLASP Conference on Learning with Small Data (LSD 2023)*.
10. **Discontinuous Constituency and BERT: A Case Study of Dutch**
Konstantinos Kogkalidis and Gijs Wijnholds
 In *Findings of the Association for Computational Linguistics (ACL 2023 Findings)*.
11. **Diamonds are Forever: Theoretical and Empirical Support for a Dependency-Enhanced Type Logic**
 Michael Moortgat, Konstantinos Kogkalidis and Gijs Wijnholds
 In *Logic and Algorithms in Computational Linguistics 2021 (LACompLing2021)*.
12. **A Logic-Based Framework for Natural Language Inference in Dutch**
 Lasha Abzianizde and Konstantinos Kogkalidis
 In *Journal of Computational Linguistics in the Netherlands (CLIN 2021)*.
13. **Fighting the COVID-19 Infodemic with a Holistic BERT Ensemble**
 Georgios Tziafas, Konstantinos Kogkalidis and Tommaso Caselli
 In *Proceedings of the 4th Workshop on NLP for Internet Freedom (NLP4IF 2021)*.
14. **Neural Proof Nets**
Konstantinos Kogkalidis, Michael Moortgat and Richard Moot
 In *Proceedings of the 24th Conference on Computational Natural Language Learning (CoNLL 2020)*.
15. **Æthel: Automatically Extracted Typological Derivations for Dutch**
Konstantinos Kogkalidis, Michael Moortgat and Richard Moot
 In *Proceedings of the 12th Language Resources and Evaluation Conference (LREC 2020)*.
16. **Deductive Parsing with an Unbounded Type Lexicon**
Konstantinos Kogkalidis, Michael Moortgat, Richard Moot and Georgios Tziafas
 In *Preprint*. (2019).
17. **Constructive Type-Logical Supertagging with Self-Attention Networks**
Konstantinos Kogkalidis, Michael Moortgat and Tejaswini Deoskar
 In *Proceedings of the 4th Workshop on Representation Learning for NLP (REPL4NLP 2019)*.
18. **Towards a 2-Multiple Context-Free Grammar for the 3-Dimensional Dyck Language**
Konstantinos Kogkalidis and Orestis Melkonian
 In *At the Intersection of Language, Logic and Information (ESSLLI 2018 Student Session)*.

Presentations (selection)

1. [Invited] **Grammaticality as Provability**
 At *The Logic Seminar of the University of Groningen*, Groningen, 2025.
2. [Contributed] **On Tables with Numbers, with numbers**
 At *The NAACL 2025 Workshop on Language Models for Underserved Communities*, Online, 2025.

3. [Invited] **LLM-free Representation Learning from Theorem Structure**
At *Deep Learning Models for Mathematics and Type Theory*, Gothenburg, 2025.
4. [Contributed] **Structure-Aware Neural Representations of Agda Programs**
At *EuroProofNet-WG5: Theorem Proving and Machine Learning in the Age of LLMs*, Edinburgh, 2025.
5. [Contributed (Poster, Spotlight)] **Algebraic Positional Encodings**
At *Neural Information Processing Systems*, Vancouver, 2024.
6. [Contributed (Poster)] **Learning Structure-Aware Representations of Dependent Types**
At *Neural Information Processing Systems*, Vancouver, 2024.
7. [Invited] **Tables with Numbers, and What They (Do Not) Mean**
At *Computational Linguistics and Language Technology Lab (CLLT) at the University of Crete*, Online, 2024.
8. [Invited] **Empiricism in Computational Linguistics**
At *Circolo Linguistico dell'Università di Bologna*, Bologna, 2024.
9. [Contributed (Poster)] **Nominal Class Assignment in Swahili: A Computational Account**
At *Computational Linguistics in Italy*, Pisa, 2024.
10. [Contributed] **Geometry-Aware Supertagging with Heterogeneous Dynamic Convolutions**
At *The 2023 CLASP Conference on Learning with Small Data*, Gothenburg, 2023.
11. [Contributed (Poster)] **Improving BERT Pretraining with Syntactic Supervision**
At *The 2023 CLASP Conference on Learning with Small Data*, Gothenburg, 2023.
12. [Contributed] **Neurosymbolic Proof Search for Linguistics**
At *Logic for the AI Spring, Lake Como School of Advanced Studies*, Como, 2023.
13. [Invited] **The Unicorn of Constant-Time Parsing**
At *End-to-End Compositional Models of Vector-Based Semantics, ESSLLI*, Galway, 2022.
14. [Lecture] **Logic-Based Parsing with Graph Neural Networks**
At *End-to-End Compositional Models of Vector-Based Semantics, ESSLLI*, Galway, 2022.
15. [Contributed] **Supertagging Beyond Trees with Heterogeneous Dynamic Convolutions**
At *Computational Linguistics in the Netherlands*, Tilburg, 2022.
16. [Internal (Demo)] **From Text to Meaning: A Type-Driven Treebank, and Applications**
At *Language Sciences Day*, Utrecht, 2021.
17. [Contributed] **Neural Proof Nets**
At *The 24th Conference on Computational Natural Language Learning*, Online, 2020.
18. [Contributed] **Deductive Parsing with an Unbounded Type Lexicon**
At *SemSpace 4, ESSLLI*, Riga, 2019.
19. [Contributed (Poster)] **Constructive Type-Logical Supertagging with Self-Attention Networks**
At *The 4th Workshop on Representation Learning for NLP*, Florence, 2019.
20. [Contributed] **From Raw Text to Linear Lambda Terms**
At *Compositionality in Formal and Distributional Models of Natural Language Semantics*, Utrecht, 2019.
21. [Contributed] **Towards a 2-MCFG for the 3-Dimensional Dyck Language**
At *ESSLLI Student Session*, Sofia, 2018.

Professional Service

ACL ARR

Reviewer (2021 -); Area Chair (2025 -)

NeurIPS

Reviewer (2025 -)

Extracurricular

Languages

Greek [native], English [fluent], Italian [intermediate], German [elementary], Dutch [elementary]